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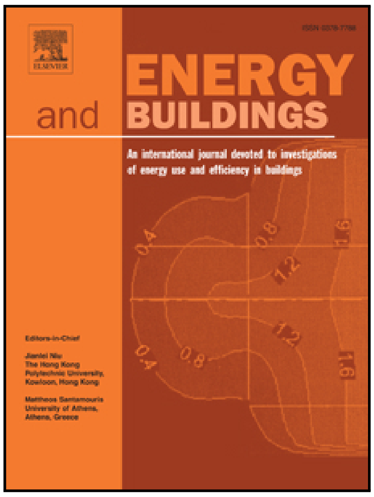

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PII: S0378-7788(26)00705-X
DOI: https://doi.org/10.1016/j.enbuild.2026.117645
Reference: ENB 117645

To appear in: *Energy & Buildings*

Received date: 28 February 2026
Revised date: 4 May 2026
Accepted date: 12 May 2026

Please cite this article as: Yan Gu, Huiru Yan, Qingle Wang, Cong Liu, Cheng Liu, Long Cheng, Quantum Reinforcement Learning for Dynamic Multi-Objective Energy Management in Community Microgrids, *Energy & Buildings* (2026), doi: https://doi.org/10.1016/j.enbuild.2026.117645

Quantum Reinforcement Learning for Dynamic Multi-Objective Energy Management in Community Microgrids

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ARTICLE INFO

Keywords:
Quantum reinforcement learning
Community microgrids
Multi-objective optimization
Smart energy systems

ABSTRACT

Deep reinforcement learning (DRL) has gained considerable attention for improving the performance of Energy Management Systems (EMSs) in microgrids, enabling more effective handling of the trade-offs among energy efficiency, operational cost, and user comfort. Simultaneously, with the rise of quantum computing, Quantum Neural Networks (QNNs) have demonstrated their unique capabilities in information processing and representation. In line with these technological advancements, this work investigates the application of Quantum Reinforcement Learning (QRL) to energy management in community microgrids. Specifically, we develop Q-EMS, a QRL-based framework that exploits the information encoding capability of quantum bits (qubits) and the function approximation capability of QNNs to support effective energy management decisions. Our approach is tailored to dynamically optimize energy utilization, enhance user comfort, and improve energy profitability. We provide a detailed design of our approach, and our results show that Q-EMS either matches or exceeds the performance of state-of-the-art DRL methods.

1. Introduction

The global energy demand has been increasing at an unprecedented rate, primarily driven by rapid industrialization, urbanization, and an ongoing dependence on fossil fuels. This trend has led to a significant increase in greenhouse gas emissions from conventional power generation methods, exacerbating environmental concerns. As a solution to these critical challenges, many countries are adopting to microgrids powered by renewable energy sources (RESs) such as solar and wind [25]. However, the variable nature of RESs introduces challenges in their reliability, often leading to inconsistent energy supply that may not always align with fluctuating requirements. To counteract these challenges, microgrids are increasingly employing sophisticated Energy Management Systems (EMSs) to optimize the production, distribution, and consumption of energy. Specifically, within a small-scale power grid, the successful implementation of an efficient EMS not only benefits consumers and utility companies financially but also leads to substantial energy savings, enhanced comfort and convenience, reduced greenhouse gas emissions, and a strengthened reliance on Energy Storage Systems (ESSs) and RESs [7, 41].

In practice, modern community microgrids operate as tightly coupled systems composed of heterogeneous components, including renewable generation units, ESSs, flexible residential loads, and bidirectional interactions with the main grid. As illustrated in Figure 1, their operation requires coordinated management of energy flows, control actions, and information exchange across multiple subsystems, which substantially increases system complexity. Moreover, operational decisions are strongly interdependent over time, as actions taken at one time step can influence future system states and affect both economic performance and user comfort. Early studies have applied a variety of optimization techniques to microgrid energy management, such as linear programming, mixed-integer nonlinear programming, and heuristic algorithms [27]. While effective in improving resource utilization and system reliability, these methods become less suitable as microgrids evolve toward community-level systems with increased scale and coupling. Their reliance on predefined rules or detailed system models, together with significant computational overhead, limits their adaptability under dynamic operating conditions. Consequently, there is a growing need for adaptable algorithms capable of dynamically responding to fluctuating energy demands and environmental changes, thereby optimizing performance in a rapidly changing energy landscape [14].

To effectively address these challenges, deep reinforcement learning (DRL) has emerged as a key enabling technology for advanced EMSs. Unlike conventional methods, DRL can autonomously learn and continuously refine energy management strategies through continuous interaction with complex and uncertain environments [12, 9], which makes it particularly well suited for EMS applications. Driven by the rapid development of quantum machine learning, Quantum Reinforcement Learning (QRL) has recently attracted increasing attention as a promising paradigm for complex control and optimization in energy management [32, 13].

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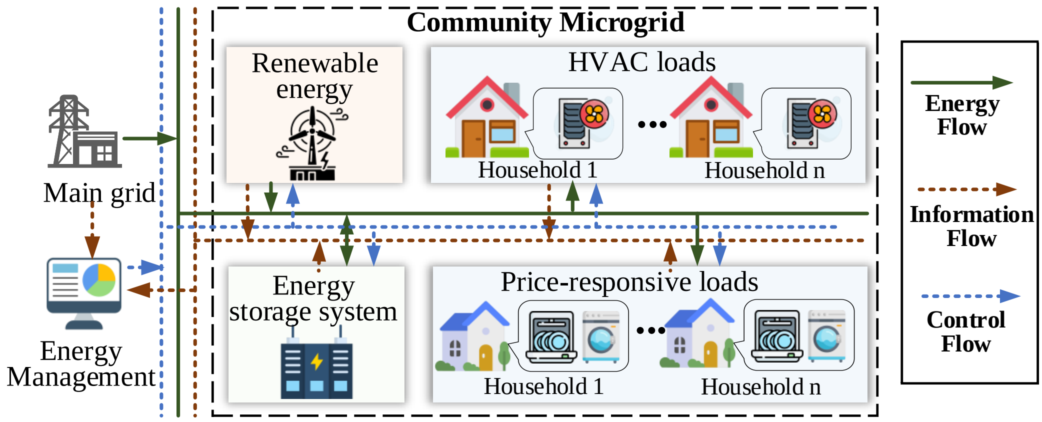


Figure 1: Overall architecture of the community microgrid energy management system.

In particular, QRL offers several potential advantages. First, by exploiting quantum mechanical properties such as superposition and entanglement, quantum bits (qubits) provide a compact and expressive way to encode system states, thereby facilitating the representation of complex state information in community microgrids. Second, trainable variational quantum layers, constructed using parameterized rotation gates and fixed entangling operations, can serve as flexible function approximators for estimating state-action values. These characteristics make QRL a promising direction for improving the learning efficiency and decision-making capability of community microgrid energy management [18].

Building on these potentials, we propose Q-EMS, a QRL framework tailored for community microgrid energy management. Leveraging the encoding capability of qubits and the expressive efficiency of Variational Quantum Circuits (VQCs), Q-EMS is designed to capture intricate decision-making patterns and to enable effective action-value estimation under dynamic operating conditions. The proposed approach dynamically optimizes energy utilization, enhances user comfort, and improves operational profitability. The main contributions of this paper are summarized as follows:

- We have introduced Q-EMS, a cutting-edge QRL framework for optimizing energy management in community microgrids. Specifically, Q-EMS incorporates a multiple re-uploading encoding strategy to enhance the expressiveness and policy learning capabilities of the QNN.
- We have incorporated renewable energy integration modules, ESSs, main grid, thermostatically controllable loads, and price-responsive loads into our microgrid environment to support dynamic energy management, with Q-EMS designed to optimize operational profit and user comfort.
- Our experiments show that Q-EMS performs as well as or better than the latest DRL method. This highlights the practical potential of Q-EMS for real-world energy management, and also demonstrates the groundbreaking impact of quantum computing solutions in this domain.

The rest of this paper is organized as follows: Section 2 reviews the relevant literature. Section 3 introduces the system model and problem formulation. The detailed design of the proposed Q-EMS is presented in Section 4, followed by experimental evaluations in Section 5. Finally, Section 6 concludes the paper.

2. Related Work

This section reviews representative studies on EMS, with a particular focus on conventional approaches and DRL–based methods. In addition, recent advances in QML are briefly discussed.

2.1. Energy Management System

An effective EMS strategy is essential for coordinating power generation, energy storage operation, and electricity consumption, thereby maintaining grid stability and economic viability [35]. Early research in community microgrid management predominantly relied on conventional techniques, which formulate energy scheduling problems using deterministic mathematical models. Among these, Mixed-integer linear programming (MILP) has been extensively applied. For example, Vilaisarn et al. [39] developed a MILP-based EMS for alternating current microgrids, incorporating hierarchical control mechanisms to handle both grid-connected and islanded operating modes. Another widely used method is Model Predictive Control (MPC), which enables dynamic, constraint-aware decision-making over a receding time horizon. Mario et al. [29] designed a hierarchical EMS that integrates optimal generation scheduling with MPC to jointly address long-term planning and short-term real-time energy dispatch in hydrogen-based microgrids. In addition, heuristic approaches have been applied to improve EMS adaptability under non-convex and uncertain conditions. For instance, Rodriguez et al. [31] proposed a fuzzy logic-based EMS for isolated microgrids, where controller parameters were optimized using Particle Swarm

Table 1
Comparison of existing energy management methods with respect to key design factors.

Ref.	Price-Responsive Loads	Renewable Power	Community Level	Indoor Comfort	Methods
Vilaisarn et al. [39]	✗	✓	✗	✗	Rule-based
Mario et al. [29]	✗	✓	✗	✗	Rule-based
Rodriguez et al. [31]	✗	✓	✓	✗	Heuristic-based
Silva et al. [38]	✗	✓	✓	✗	Heuristic-based
Gao et al. [8]	✗	✗	✗	✓	DRL-based
Yu et al. [44]	✓	✓	✗	✓	DRL-based
Ren et al. [30]	✓	✓	✗	✓	DRL-based
Zhao et al. [46]	✓	✓	✓	✓	DRL-based
Wu et al. [42]	✓	✓	✓	✗	DRL-based
This work	✓	✓	✓	✓	QRL-based

Optimization (PSO) and Cuckoo Search to enhance power coordination among photovoltaic generation (PV), diesel generators, and battery systems. Silva et al. [38] proposed a hybrid ant colony optimization and genetic algorithm for energy management in smart communities, achieving reduced electricity costs and improved renewable energy utilization while maintaining user comfort. Despite their effectiveness, these conventional optimization and heuristic-based approaches typically rely on accurate system models and often incur substantial computational overhead, which limits their applicability in real-time operation and highly dynamic energy management scenarios [40].

To overcome the inherent limitations of conventional energy management strategies, DRL has been increasingly investigated as an alternative paradigm for community microgrids. By learning control policies through continuous interaction with the environment, DRL alleviates the dependence on explicitly formulated system models and exhibits strong adaptability to dynamic behaviors and operational uncertainties. For example, Gao et al. [8] designed a Deep Deterministic Policy Gradient (DDPG)-based control strategy for building HVAC systems to jointly reduce energy consumption and improve occupant comfort under time-varying thermal conditions. Yu et al. [44] further extended DDPG to smart home energy management, enabling coordinated control of HVAC units and energy storage systems without relying on explicit thermal models or prior knowledge of uncertain parameters. More recently, DRL has been scaled to community-level energy management. Ren et al. [30] developed a soft Actor-Critic-based home energy management approach that explicitly incorporates uncertainties in electricity prices, PV, and user comfort preferences, leading to more robust resource scheduling. From a low-carbon perspective, Zhao et al. [46] introduced a Dueling Double Deep Q-Network (D3QN) with a mixed penalty function to balance economic performance, carbon emissions, and user-related constraints within a unified DRL framework. Wu et al. [42] developed a DDPG-based online EMSs for microgrids with high renewable penetration, targeting multi-objective optimization under strong uncertainties arising from renewable generation, load dynamics, and flexible electricity markets.

Collectively, these studies highlight the effectiveness of DRL for complex energy management. Building on these advances and recent progress in quantum computing, this work explores QRL for community microgrid energy management by leveraging the expressive and parameter-efficient characteristics of quantum neural networks to address sequential decision-making under uncertainty.

2.2. Quantum Machine Learning

QML integrates quantum computing paradigms with machine learning techniques, offering new computational mechanisms for data representation and parallel information processing [15, 1]. From a theoretical perspective, Gil-Fuster et al. [10] showed that QNNs exhibit considerable expressive capacity even in finite-sample regimes, enabling them to fit complex learning patterns beyond classical counterparts. Gonon et al. [11] established universal approximation results with explicit error bounds for both trainable and randomly initialized QNNs, showing that a broad class of target functions can be approximated with polynomial sample complexity and only logarithmic scaling in the number of qubits, without suffering from the curse of dimensionality. Motivated by these advances, QNNs have recently been investigated in power and energy applications. For example, Kumar et al. [20] proposed a QNN-based power distribution and supply management model that estimates regional smart-meter electricity demand based on load forecasting results. Zhu et al. [47] developed a Bayesian QNN-based power flow calculation method to characterize the impact of renewable generation uncertainty on power flow states. Although these studies demonstrate the potential of QNNs for energy system modeling, relatively few QML-based studies have explored their use in sequential decision-making for energy management. This gap is particularly important for community microgrids, where energy management requires adaptive control decisions under stochastic renewable generation, time-varying electricity prices, and heterogeneous household demand.

In parallel, QRL has emerged as a promising extension of QML for sequential decision-making problems. Some studies have demonstrated that quantum mechanisms can enhance learning efficiency. For example, Saggio et al. [34] demonstrated that quantum communication channels can

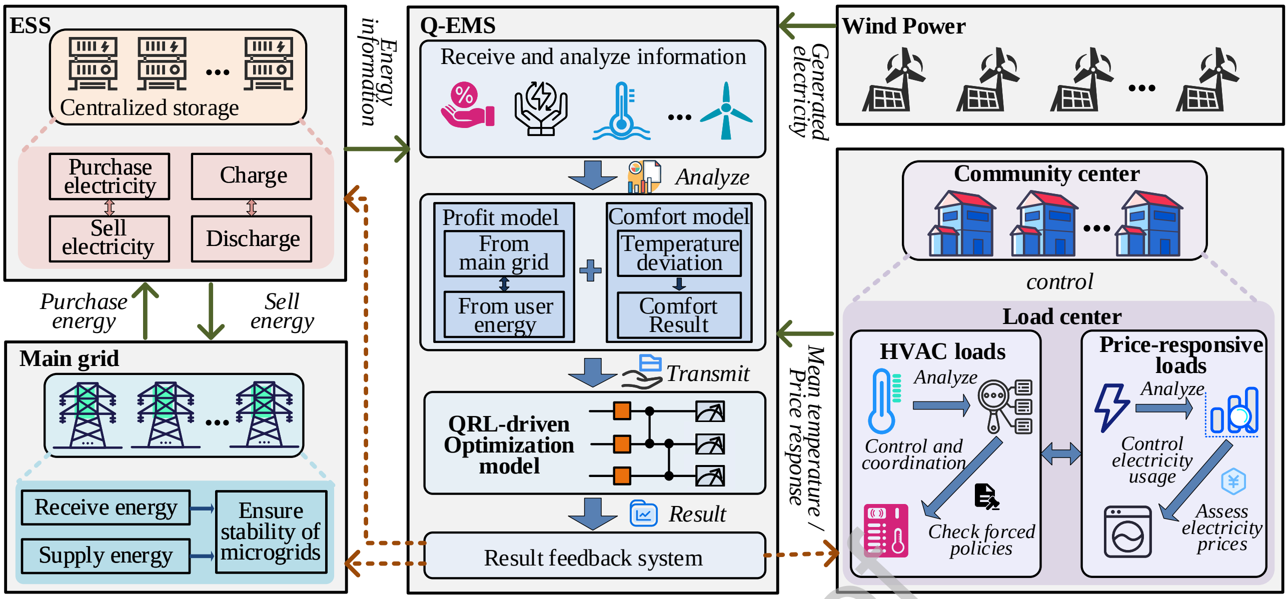


Figure 2: The framework of community microgrid energy management using Q-EMS.

accelerate reinforcement learning by enabling more efficient information exchange during training. Building upon this direction, Yun et al. [45] integrated QNNs into a multi-agent reinforcement learning framework, improving learning efficiency and adaptability in cooperative settings. Shi et al. [37] investigated energy management for multi-stack fuel cell vehicles using a hybrid QRL framework, where parameterized quantum circuits were embedded into the actor network of a DDPG controller, enabling competitive control performance with substantially fewer trainable parameters. In a different application domain, Kim et al. [19] formulated the stabilized landing of reusable rockets as a reinforcement learning problem and employed a parameterized QNN as the policy function approximator, mapping system states to control actions through quantum state encoding, variational quantum circuits, and measurement. Despite these promising results, the application of QRL to energy management in community microgrids remains largely underexplored. Motivated by this gap, this work develops a QRL framework for energy management in community microgrids, aiming to address the challenges arising from complex system dynamics and long-term decision dependencies.

Brief summary. Based on the comparative analysis in Table 1, most existing energy management approaches predominantly rely on heuristic optimization techniques or DRL–based control methods. Among these, DRL-based approaches have attracted increasing attention due to their ability to learn effective decision-making policies through continuous interaction with dynamic environments, thereby exhibiting strong adaptability in complex operational scenarios. Motivated by the potential of quantum computing to enhance expressive state representations and decision efficiency, this work investigates a QRL paradigm for community microgrid energy management. The proposed framework explicitly incorporates price-responsive loads, renewable energy integration, coordination among multiple users, and indoor comfort, offering a unified and flexible solution for managing the complex operational characteristics of community microgrids.

3. System Model and Problem Formulation

This section describes the system architecture and formulates the community energy management problem addressed in this work. Key symbols and their definitions are provided in Table 2.

3.1. Overall Architecture

This work focuses on the energy management of a community microgrid that integrates renewable energy sources, centralized energy storage, and heterogeneous residential loads. As illustrated in Figure 2, the community microgrid comprises wind turbines for renewable power generation, a centralized ESS, and multiple residential households with diverse energy consumption patterns. At the household level, two primary categories of loads are considered: TCL loads, and price-responsive loads that follow daily consumption patterns. The community microgrid operates in a grid-connected mode and participates in the electricity market to maintain energy balance through power exchange with the main grid. The proposed Q-EMS coordinates ESS power dispatch and household load control at each discrete time interval t, while accounting for user comfort and economic performance. From the operator's perspective, the objective is to minimize operational costs and maximize economic benefits. In this work, a Time-of-Use (ToU) tariff scheme

Table 2
Notations used in Q-EMS system.

Symbol	Description	Symbol	Description
c_t^{price}	Electricity price for price-responsive loads [cent/kWh]	c_{gen}	Unit cost of renewable power generation . . [cent/kWh]
$cost_t$	Total operating cost . . . . [cent]	c_{tcl}	Electricity price for TCL loads . . . . . . [cent/kWh]
E_t	Stored energy of the ESS at time t . . . . . . [kWh]	E_t^{SoC}	State of charge of the ESS . . . . [-]
$E_{\min}$	Minimum ESS energy capacity . . . . [kWh]	$E_{\max}$	Maximum ESS energy capacity . . . . [kWh]
h_i	i-th residential household . . . . [-]	$L_{i,t}^{pr}$	Price-induced load adjustment . . . . [kw]
$L_{i,t}^{re}$	Recovered deferred load at time t . . . . [kw]	$L_{i,t}^{de}$	Electricity demand of household h_i . . . . [kw]
P_t^{grid}	Power exchanged with the main grid . . . . . [kw]	P_t^{char}	ESS charging power . . . . [kw]
$P_{t,max}^{char}$	Maximum ESS charging power . . . . [kw]	P_t^{dis}	ESS discharging power . . . . [kw]
$P_{t,max}^{dis}$	Maximum ESS discharging power . . . . . . [kw]	P_t^{ac}	Aggregate TCL power consumption . . . . [kw]
p_i^{ac}	Rated TCL power of household h_i . . . . . . [kw]	P_t^{price}	Aggregate price-responsive load demand . . . . [kw]
P_t^{rew}	Renewable power generation . . . . [kw]	P_t^{buy}	Power imported from the main grid . . . . [kw]
p_t^{com}	Community-level temperature deviation . . . . [°C]	P_t^{sell}	Power exported to the main grid . . . . [kw]
r_t^c	Community-level comfort metric . . . . [-]	r_t^p	Operational profit . . . . [cent]
rev_t	Total operational revenue . . . . [cent]	$T_t^{avg\_tcl}$	Average normalized indoor temperature . . . . . . [-]
T_i^t	Indoor air temperature of household h_i . . . . [°C]	$T_{\min}$	Lower comfort temperature bound . . . . [°C]
$T_{\max}$	Upper comfort temperature bound . . . . . . [°C]	u_t	Electricity purchase price from the main grid [cent/kWh]
u_i^t	Actual TCL operating action . . . . [-]	$\hat{u}_i^t$	TCL control action . . . . [-]
v_t	Electricity selling price to the main grid . [cent/kWh]	V_t	Binary ESS operation mode . . . . [-]
η_c	Charging efficiency of the ESS . . . . [-]	η_d	Discharging efficiency of the ESS . . . . [-]
δ_t	Normalized electricity price signal . . . . [-]	ω	Maximum comfort score . . . . [-]
β_i	Price sensitivity coefficient of household h_i . . . [-]	$\mathcal{K}_i^t$	Set of deferred load indices . . . . [-]
ΔT	Temperature margin for comfort extension . . . [°C]	$\pi_i^{k,t}$	Recovery probability of deferred load k . . . . . . [-]

Table 3
Time-of-Use electricity prices adopted in this work.

Period	Time	Purchase Price (cent/kWh)	Sell Price (cent/kWh)
Normal hours	8:00-12:00, 13:00-19:00	4.5	3.8
Peak hours	12:00-15:00, 19:00-22:00	6.5	4.9
Valley hours	0:00-8:00, 22:00-24:00	3.0	2.7

is adopted [21], and the corresponding electricity purchase and selling prices are summarized in Table 3.

3.2. System Model

The community microgrid considered in this work comprises n residential households, denoted by $\{h_i\}_{i=1}^n$, and is modeled at the community level. Each household includes TCL loads and price-responsive loads, capturing both comfort-sensitive and flexible electricity demand. At the community level, the system integrates a centralized ESS, wind power generation units, and a connection to the main grid, enabling coordinated energy management among different system components. The mathematical models of the ESS, renewable generation, household loads, and grid interaction are presented in the following.

3.2.1. Energy Storage Systems

The ESS plays a pivotal role in the community microgrid by providing flexibility for power balancing and economic optimization. Its charging and discharging behavior is governed by the energy management strategy and constrained by operational limits, including energy capacity and power-rate constraints. By regulating its power exchange with the microgrid, the ESS supports efficient energy utilization and improved economic performance. The corresponding ESS dynamics at each time step t are modeled as follows:

$$\begin{aligned}
&E_t = E_{t-1} + \eta_c P_t^{char}\Delta t - \frac{P_t^{dis}}{\eta_d}\Delta t\\
&E_t^{SoC} = E_t / E_{\max}\\
&\text{s.t.}\quad \text{c1.}\quad E_{\min} \le E_t \le E_{\max}\\
&\qquad\quad \text{c2.}\quad 0 \le P_t^{char} \le P_{t,\max}^{char} V_t\\
&\qquad\quad \text{c3.}\quad 0 \le P_t^{dis} \le P_{t,\max}^{dis}(1 - V_t)
\end{aligned} \tag{1}$$

where E_t represents the energy stored in the ESS at time step t, η_c and η_d denote the charging and discharging efficiency coefficient, respectively. P_t^{char} and P_t^{dis} represent the charging and discharging power of the ESS, which are constrained by their corresponding maximum limits $[0, P_{t,max}^{char}]$ and $[0, P_{t,max}^{dis}]$. E_t^{SoC} is the state of charge (SoC) of the ESS.

Constraint c1 enforces the minimum and maximum energy capacity limits of the ESS. Constraints c2 and c3 introduce a binary control variable V_t to prevent simultaneous charging and discharging operations. Specifically, $V_t = 1$ indicates that the ESS operates in charging mode, allowing $P_t^{char} > 0$ while forcing $P_t^{dis} = 0$, whereas $V_t = 0$ corresponds to the discharging mode, in which $P_t^{dis} > 0$ and $P_t^{char} = 0$.

3.2.2. Thermostatically Controllable Loads

Thermostatically controllable loads (TCLs) are employed to regulate indoor temperature and maintain thermal comfort in residential households. Each TCL operates in two discrete modes, namely active ("ON") and inactive ("OFF"), which are represented by 1 and 0, respectively. In our system, TCLs are modeled as heating-mode thermostatic loads, where the ON state increases the indoor temperature and the OFF state stops the heating contribution. These TCL units can be directly controlled at each time step t through control signals issued by the TCL aggregator. Let $\hat{u}_i^t \in \{0, 1\}$ denote the operating state assigned to the i-th TCL at time step t. To maintain indoor temperature within the extended comfort band under the heating-mode TCL model, an override control mechanism is introduced. Specifically, when the indoor temperature falls below the lower bound, the TCL is forced to the ON state to prevent excessively low indoor temperatures. When the indoor temperature exceeds the upper bound, the TCL is forced to the OFF state to prevent further heating. Within the extended comfort band, the assigned control action is retained. Accordingly, the actual control signal after override is denoted by u_i^t, which is defined as

$$u_i^t = \begin{cases} 0, & \text{if } T_i^t > T_{\max} + \Delta T \\ \hat{u}_i^t, & \text{if } T_{\min} - \Delta T \le T_i^t \le T_{\max} + \Delta T \\ 1, & \text{if } T_i^t < T_{\min} - \Delta T \end{cases} \tag{2}$$

where ΔT denotes a predefined temperature margin used to extend the nominal comfort bounds.

To characterize the dynamic interaction between indoor temperature evolution and TCL control actions, a second-order equivalent thermal parameter (ETP) model is adopted [22]. The corresponding thermal dynamics are formulated as:

$$\begin{aligned} T_i^{t+1} &= T_i^t + \frac{1}{c_a^i}\left(T_{out}^t - T_i^t\right) + \frac{1}{c_m^i}\left(T_{m,i}^t - T_i^t\right) + p_i^{ac} u_i^t + q_i \\ T_m^{t+1} &= T_{m,i}^t + \frac{1}{c_m^i}\left(T_i^t - T_{m,i}^t\right) \end{aligned} \tag{3}$$

where T_i^t denotes the indoor air temperature of household h_i at time step t, and T_{out}^t represents the outdoor temperature. The variable $T_{m,i}^t$ denotes the building thermal mass temperature, which captures the heat storage effect of the building envelope. The parameters c_a^i and c_m^i represent the effective thermal masses of the indoor air and building materials, respectively. The term q_i denotes the internal heat gain of the household, while p_i^{ac} denotes the nominal electrical power consumption of the i-th TCL unit in the ON state. Therefore, based on the actual operating states of all TCL units, the aggregate TCL power consumption at time step t is calculated as $P_t^{ac} = \sum_{i=1}^n u_i^t p_i^{ac}$.

For each household h_i, the indoor temperature T_i^t is used to characterize its thermal comfort condition at time step t. To describe the overall thermal status of the community, an aggregate comfort indicator is defined by averaging the normalized indoor temperatures of all households, as follows:

$$T_t^{avg\_tcl} = \frac{1}{n}\sum_{i=1}^n \frac{T_i^t - T_{\min}}{T_{\max} - T_{\min}} \tag{4}$$

where $T_t^{avg\_tcl}$ denotes the average normalized indoor temperature across the community, reflecting the relative thermal condition of residential households.

3.2.3. Price-Responsive Loads

Price-responsive loads refer to residential appliances whose operation can be flexibly adjusted in response to electricity price signals, such as washing machines and dishwashers. By temporarily reducing electricity consumption during high-price periods and shifting the deferred demand to more favorable time intervals, these loads provide additional demand-side flexibility at the household level. Such load-shifting behavior contributes to peak demand mitigation and can effectively reduce electricity expenditure without violating appliance operational requirements. The electricity demand of household h_i at time step t is denoted by $L_{i,t}^{de}$. The amount of demand adjusted in response to the electricity price signal at time step t is modeled as:

$$L_{i,t}^{pr} = \beta_i \cdot \delta_t \cdot L_{i,t}^{de} \tag{5}$$

where $\beta_i \in [0, 1]$ represents the price sensitivity coefficient of household h_i, and δ_t denotes the normalized electricity price level at time step t. A positive value of $L_{i,t}^{pr}$ indicates load curtailment and deferral under high-price conditions, whereas a negative value corresponds to load advancement when electricity prices are relatively low.

The curtailed load is not permanently discarded but accumulated as deferred demand to be recovered in subsequent time steps. For each household h_i, the set $\mathcal{K}_i^t = \{\, k < t \mid L_{i,k}^{pr} > 0 \,\}$ collects the time indices of deferred load segments generated prior to time step t that have not yet been recovered. The amount of deferred load recovered at time step t is modeled as:

$$\begin{aligned} L_{i,t}^{re} &= \sum_{k \in \mathcal{K}_i^t} \pi_i^{k,t} L_{i,k}^{pr} \\ P_t^{price} &= \sum_{i=1}^n \left(L_{i,t}^{de} - L_{i,t}^{pr} + L_{i,t}^{re}\right) \end{aligned} \tag{6}$$

where the recovery process is modeled probabilistically, and $\pi_i^{k,t}$ denotes the likelihood that a deferred load segment originating at time step k is executed at time step t. The term P_t^{price} represents the aggregate power demand contributed by price-responsive loads within the community at t.

3.2.4. Power Balance

The main grid serves as a key component in supporting reliable microgrid operation by compensating for deviations between electricity generation and consumption. To ensure secure and feasible operation, the microgrid must satisfy an instantaneous power balance condition at every time step. Accordingly, the power balance constraint at time step t is formulated as:

$$P_t^{rew} + P_t^{dis} + P_t^{grid} = P_t^{ac} + P_t^{price} + P_t^{char} \tag{7}$$

where the left side represents the power supply of the microgrid, P_t^{rew} denotes the power generated by renewable energy sources, such as wind turbines, and P_t^{dis} represents the discharging power provided by the ESS. P_t^{grid} denotes the power exchanged between the microgrid and the main grid. A positive value of P_t^{grid} indicates that electricity is imported from the main grid to support microgrid operation, whereas a negative value indicates that surplus electricity is exported from the microgrid to the main grid. On the demand side, P_t^{ac} denotes the aggregated power consumption of TCL loads, P_t^{price} represents the electricity demand of price-responsive loads, and P_t^{char} denotes the charging power absorbed by the ESS.

3.3. Optimization Objectives

The proposed Q-EMS aims to jointly improve the economic performance of the community microgrid and the thermal comfort of end users over a time horizon of T discrete time steps. Specifically, the optimization seeks to maximize the expected operational profit while ensuring that user comfort remains within acceptable bounds, subject to power balance and operational constraints. The corresponding optimization objectives are formulated as follows.

3.3.1. Economic Profit

The economic performance of the community microgrid at each interval t is evaluated through the operational profit, denoted by r_t^p, which is defined as the difference between the total revenue rev_t and the total operating cost $cost_t$ incurred during that interval. Specifically, r_t^p is expressed as:

$$\begin{aligned}
r_t^p &= rev_t - cost_t \\
rev_t &= c_t^{price} P_t^{price} + c_{tcl} P_t^{ac} + v_t P_t^{sell} \\
cost_t &= c_{gen} P_t^{rew} + u_t P_t^{buy}
\end{aligned} \tag{8}$$

where rev_t represents the total revenue obtained at time step t, including income from price-responsive loads, TCL loads, and the sale of surplus electricity to the main grid. The operating cost $cost_t$ consists of the generation cost associated with local renewable resources and the expenditure incurred when purchasing electricity from the main grid. In this formulation, c_t^{price}, c_{tcl}, and v_t denote the unit electricity prices applied to price-responsive loads, TCL loads, and electricity exported to the main grid, respectively. Meanwhile, c_{gen} represents the estimated levelized cost of wind energy, and u_t denotes the electricity purchase price from the main grid at time step t. P_t^{sell} and P_t^{buy} denote the amount of electricity sold to and purchased from the main grid at time step t, respectively.

3.3.2. Comfort

The primary role of the thermostatically controllable system is to regulate indoor temperatures within a predefined comfort range, which is essential for maintaining residents' thermal satisfaction. To assess thermal comfort at the community level, an aggregated temperature deviation metric, denoted by p_t^{com}, is introduced to quantify the overall violation of comfort bounds across all households at time step t. Based on this aggregated deviation, a community-level comfort evaluation metric r_t^c is defined as:

$$r_t^c = \begin{cases} \omega, & \text{if } p_t^{com} = 0, \\ -\omega\left(\dfrac{p_t^{com}}{\kappa}\right)^3, & \text{if } 0 < p_t^{com} \le \kappa, \\ -\omega, & \text{if } p_t^{com} > \kappa, \end{cases}$$
$$p_t^{com} = \sum_{i=1}^{n} \Big(\max(T_i^t - T_{\max}, 0) + \max(T_{\min} - T_i^t, 0) \Big) \tag{9}$$

where T_i^t denotes the indoor temperature of household h_i at time step t, while $T_{\min}$ and $T_{\max}$ denote the lower and upper bounds of the acceptable comfort range, respectively. When $p_t^{com} = 0$, all households remain within the comfort range and a positive reward is assigned. Once $p_t^{com} > 0$, thermal discomfort occurs and the reward decreases monotonically with the aggregated deviation. When p_t^{com} reaches κ, the reward saturates at $-\omega$ and remains unchanged for larger violations. This formulation rewards perfect comfort preservation while imposing increasingly severe penalties on comfort violations.

3.4. Markov Decision Process for Q-EMS

The energy management problem is formulated as a MDP, represented by a five-tuple $\langle S, A, R, p, \gamma \rangle$, where S denotes the system state, A the control action, R the reward function, p the state transition probability, and γ the discount factor. At each time step t, the agent observes the current state S_t of the microgrid and selects an action A_t according to its policy. Following the execution of A_t, the system transitions to a new state S_{t+1} and an immediate reward R_t is obtained. This decision-making process proceeds over successive decision stages. The objective of the agent is to learn an optimal policy that maximizes the expected cumulative discounted reward, thereby enabling efficient and adaptive energy management under uncertain operating conditions. To solve this MDP, the proposed Q-EMS employs a QRL framework, where a QNN is used to approximate the action-value function, as detailed in Section 4. The state, action, and reward components are described as follows.

3.4.1. State Space

The system state at time step t, denoted by S_t, characterizes the operating condition of the community microgrid together with the external factors relevant to energy management. This state vector is subsequently encoded into a quantum state, as detailed in Section 4.3. Formally, the state is defined as:

$$S_t = (T_{out}^t, T_t^{avg\_tcl}, L_t^{de}, t, E_t^{SoC}, P_t^{rew}, u_t) \tag{10}$$

where T_{out}^t represents the outdoor temperature, $T_t^{avg\_tcl}$ the average indoor temperature across the community, L_t^{de} the community electricity demand load, t the time step, E_t^{SoC} the state of charge of the ESS, P_t^{rew} the renewable power generation, and u_t the electricity purchase price from the main grid.

3.4.2. Action Space

The action space of the proposed Q-EMS is formulated as a finite composite decision space that jointly coordinates TCL operation, price-responsive load regulation, and ESS operation. Based on the Q-values estimated by the QNN, the agent selects an action from this composite action space, as detailed in Section 4.4. At each decision step t, the selected composite action is defined as

$$A_t = (a_t^{tcl},\ a_t^{price},\ a_t^{ess}) \tag{11}$$

where a_t^{tcl}, a_t^{price}, and a_t^{ess} denote the TCL operation decision, the price-responsive load regulation decision, and ESS operation decision, respectively. With 4, 3, and 3 discrete choices for these components, the joint action space contains 36 composite actions.

The three action components are further defined as follows. The TCL action $a_t^{tcl} \in \{0, 1, 2, 3\}$ specifies a discrete aggregate power level for all TCL units. Based on this aggregate control level, individual TCL units are activated according to their thermal priorities, where households with lower indoor temperatures are preferentially served to maintain thermal comfort. The price-responsive load action $a_t^{price} \in \{0, 1, 2\}$ determines the discrete electricity price signal for flexible loads, which affects their consumption patterns and supports demand shifting across time periods. The energy storage management action $a_t^{ess} \in \{0, 1, 2\}$ determines the operating mode of the energy storage system, corresponding to charging, idle operation, and discharging at predefined power levels. When charging or discharging is selected, the ESS operates at the corresponding power level while satisfying its capacity and power constraints.

3.4.3. Reward Function

The reward function R_t defines the immediate benefit or penalty the system receives after taking an action A_t in state S_t at time t. In the proposed Q-EMS framework, the optimization objective is to maximize the cumulative reward over time, which reflects a trade-off between economic profitability r_t^p and user comfort r_t^c. The reward function can be defined as:

$$R_t = r_t^p + \mu r_t^c \tag{12}$$

where $\mu > 0$ is a tunable weighting coefficient that balances economic performance and user comfort, allowing the reward design to be adapted to different community preferences and operational requirements.

4. QRL Architecture Design for Q-EMS

Building on the above MDP formulation, this section presents the architectural design and implementation details of the proposed QRL framework for Q-EMS, including the QNN structure, quantum state encoding strategy, and training procedure.

4.1. Quantum Computing

Quantum computing operates on qubits, which constitute the fundamental units of information processing in quantum systems. A single qubit can be conveniently represented using the Bloch sphere, which provides a geometric interpretation of its quantum state and the effect of quantum operations acting upon it [23]. Unlike classical bits restricted to binary states, a qubit can reside in a coherent superposition of the basis states $|0\rangle$ and $|1\rangle$, enabling a richer state representation that is particularly suitable for parameterized quantum models [3]. Within the Bloch sphere representation, the evolution of a qubit state is realized through a sequence of rotation operations about different axes. Starting from an initial state $|0\rangle$, a general single-qubit state $|\psi\rangle$ can be obtained by applying a series of parameterized rotation gates [17, 2], expressed as:

$$|\psi\rangle = R_z(\theta_z) R_y(\theta_y) R_x(\theta_x)|0\rangle \tag{13}$$

where R_x, R_y, and R_z represent rotation operations around the x-, y-, and z- axes of the Bloch sphere, respectively. θ_x, θ_y, and θ_z representing their corresponding rotation angles.

These rotation gates form the basic building blocks of parameterized quantum circuits, as they allow continuous control over qubit states through tunable parameters [5]. Their matrix representations are given by:

$$\begin{aligned} R_x(\theta_x) &= \begin{bmatrix} \cos(\theta_x/2) & -i\sin(\theta_x/2) \\ -i\sin(\theta_x/2) & \cos(\theta_x/2) \end{bmatrix} \\ R_y(\theta_y) &= \begin{bmatrix} \cos(\theta_y/2) & -\sin(\theta_y/2) \\ \sin(\theta_y/2) & \cos(\theta_y/2) \end{bmatrix} \\ R_z(\theta_z) &= \begin{bmatrix} \exp(-i\theta_z/2) & 0 \\ 0 & \exp(i\theta_z/2) \end{bmatrix} \end{aligned} \tag{14}$$

Multi-qubit gates play a central role in quantum circuits by enabling controlled interactions among qubits, which are indispensable for the generation of entanglement. Among these operations, the controlled-Z (CZ) gate is widely adopted, as it introduces a conditional phase shift on the joint quantum state while preserving the amplitudes of the computational basis states. This characteristic makes the CZ gate particularly suitable for constructing entangled states and for enabling transformations that depend on relative phase information in parameterized quantum circuits. The unitary representation of a generalized CZ gate can be

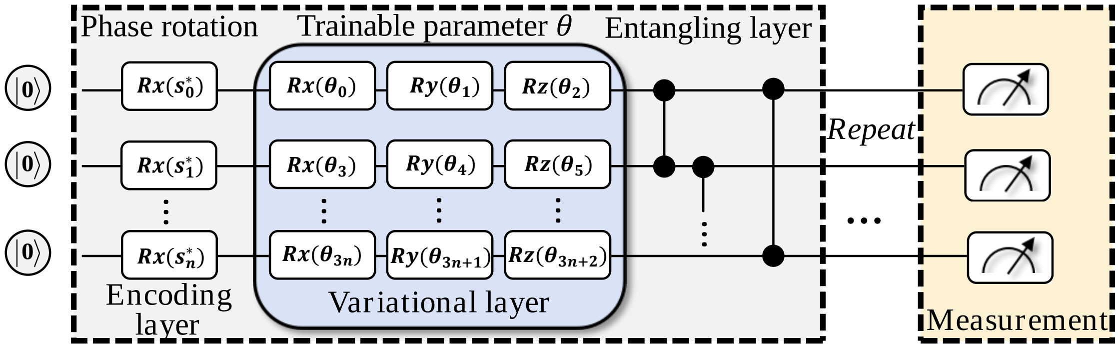


Figure 3: The quantum circuit design for Q-EMS.

expressed as:

$$U_{CZ} = e^{i\gamma} \begin{bmatrix} e^{i\alpha} & 0 & 0 & 0 \\ 0 & e^{i\beta} & 0 & 0 \\ 0 & 0 & e^{i\beta} & 0 \\ 0 & 0 & 0 & e^{i\delta} \end{bmatrix} \tag{15}$$

where α, β, and δ denote phase parameters associated with different computational basis states, and γ denotes a global phase factor. By appropriately configuring these phase parameters, the CZ gate can be employed to induce structured entanglement patterns tailored to specific computational or learning objectives.

4.2. Quantum Neural Network

A QNN is a parameterized quantum circuit equipped with trainable variables for optimization and decision-making in complex environments [24, 43]. In the proposed Q-EMS framework, the QNN is organized into three sequential stages, including state encoding (U_{enc}), variational transformation (U_{var}), and measurement (M). At time step t, the observed system state is first organized into a d-dimensional classical feature vector. In the proposed QNN, each state dimension is independently encoded into a dedicated qubit through a single-qubit rotation gate. This one-to-one encoding scheme minimizes cross-feature interference, supports low-depth state preparation, and facilitates straightforward mapping between the classical input space and the quantum Hilbert space. Accordingly, d qubits are used to encode the d-dimensional system state. Through the encoding operation U_{enc}, each classical feature is mapped onto its corresponding qubit, resulting in an initial quantum state denoted by $|\psi_{enc}\rangle$. The resulting composite quantum state can be expressed as:

$$|\psi_{enc}\rangle = |\psi_1\rangle \otimes |\psi_2\rangle \otimes \cdots \otimes |\psi_d\rangle = \sum_{k=0}^{2^d-1} \alpha_k |k\rangle \tag{16}$$

where α_k denotes the complex amplitude associated with the computational basis state $|k\rangle$. Here, $d = 7$ denotes both the dimensionality of the encoded system state and the number of qubits used in the QNN.

Following the encoding process, a parameterized unitary operator $U_{var}(\theta)$ is applied to the encoded quantum state, where θ denotes the set of trainable parameters. This variational transformation performs a sequence of rotations and entangling operations in the high-dimensional Hilbert space, enriching the representational capacity of the QNN and enabling it to capture complex dependencies critical for energy management decision-making [4]. In the final measurement stage, the expectation value of a predefined quantum observable is extracted from the resulting quantum state and mapped to a classical scalar output. This value is interpreted as the Q-value associated with the current system state and the selected control action, and is subsequently used to update the QNN parameters within the reinforcement learning process, enabling progressive refinement of the energy management policy.

4.3. Encoding Microgrid Environments into the QNN

The input state vector at time step t of the Q-EMS system is denoted as $S_t = (T_{out}^t, T_t^{avg\_tcl}, L_t^{de}, t, E_t^{SoC}, P_t^{rew}, u_t)$. To incorporate this classical state into the quantum circuit, each component of S_t is assigned to a dedicated qubit for quantum encoding. To ensure compatibility with quantum gate operations, each classical variable $s_i \in S_t$ is normalized to the interval $[-\pi, \pi]$, enabling efficient mapping to quantum rotation parameters. The resulting normalized variable is denoted by s_i^* and is computed by:

$$s_i^* = \frac{2\pi(s_i - s_{min})}{s_{max} - s_{min}} - \pi, \quad i \in \{0, 1, 2, \dots, 6\}, \quad s_i \in S_t \tag{17}$$

where s_{min} and s_{max} denote the minimum and maximum bounds of the corresponding state variable. The resulting s_i^* serves as the angular parameter for quantum rotations.

Each normalized value s_i^* is then encoded onto its corresponding qubit via a rotation around the X-axis using the Pauli-X rotation gate. The encoding operation is given by:

$$|\phi_i'\rangle = R_x(s_i^*)|\phi_i\rangle \tag{18}$$

where $|\phi_i\rangle$ denotes the initial quantum state of the i-th qubit prior to encoding, and $R_x(s_i^*)$ represents the Pauli-X rotation gate used to embed the normalized classical input s_i^* into the quantum state. After this operation, the quantum state $|\phi_i'\rangle$ embeds the normalized classical information into the corresponding qubit.

Compared with classical DRL approaches, where state information is processed through multiple layers of weighted linear transformations, the quantum encoding strategy in our system maps each state variable directly onto an individual qubit using elementary quantum gates. Such a representation avoids explicit weight expansion with increasing state dimensionality and provides a compact, structured mechanism for state embedding, which is particularly suitable for energy management problems with heterogeneous and time-varying system states [36].

4.4. Combined Layers for Multiple Re-Uploading

In the proposed Q-EMS framework, a QNN is designed to cope with the inherent complexity of community microgrid energy management. The QNN architecture integrates encoding layers and variational layers to jointly process classical system states and generate adaptive control decisions. Specifically, the encoding layers embed classical state variables into quantum states, while the variational layers, parameterized by trainable variables, progressively transform these states to approximate the desired control policy. As illustrated in Figure 3, each classical input is first encoded into the quantum register through Pauli-X gates. The resulting quantum states are then refined by parameterized single-qubit gates within the variational layers. However, these operations remain confined to individual qubits and are insufficient to capture dependencies among different state components. To address this limitation, CZ gates are introduced between adjacent qubits within each variational block. This structured entanglement couples the quantum states across the register, allowing information encoded on individual qubits to be integrated into a coherent, system-level representation. The entanglement operations are applied sequentially and can be expressed as:

$$|\Phi_{out}\rangle = (\prod_{i=1}^{6} \mathrm{CZ}_i^{i+1})|\phi_1''\rangle \otimes |\phi_2''\rangle \otimes \cdots \otimes |\phi_7''\rangle \quad (19)$$

where $|\Phi_{out}\rangle$ denotes the final entangled quantum state. Through this cascading entanglement structure, information initially encoded on separate qubits is progressively integrated across the quantum system, enabling a global representation of the microgrid state within the QNN.

Building on this architecture, the Q-EMS further adopts a multiple re-uploading strategy, in which the encoding and variational layers are combined into a composite quantum block and applied repeatedly. This mechanism allows the same classical microgrid state to be embedded into the quantum system multiple times, while intermediate variational transformations continuously adapt the quantum representation according to the learned control policy. By periodically re-encoding the input state, the QNN incrementally refines its internal quantum representation, thereby enhancing its expressive capacity and its ability to approximate complex decision functions. Formally, the output of a QNN with L re-uploading layers is given by:

$$f(s_i;\theta) = \langle 0|U_{var}^{(L)}(\theta_L)U_{enc}(s_i)\cdots U_{var}^{(1)}(\theta_1)U_{enc}(s_i)|0\rangle, \quad (20)$$

where $U_{enc}(s_i)$ denotes the unitary operator encoding the classical input state s_i, and $U_{var}^{(l)}(\theta_l)$ represents the parameterized variational unitary at the l-th layer. The quantum system is initialized in the ground state $|0\rangle$, which serves as the starting point for the iterative transformation process.

Algorithm 1: Training Procedure for Q-EMS

Initialize Q-network Q_θ and target Q-network $Q_{\theta'}$;
Initialize replay buffer $\mathcal{D}$, discount factor γ, exploration rate ϵ, and target update frequency ζ;
for *each episode* **do**
  Reset environment and observe initial state s_0 ;
  while *episode not terminated* **do**
    Normalize system state s_t to obtain s_t^* according to Equation 17;
    Encode s_t^* into quantum state using $U_{enc}(s_t^*)$;
    Evaluate action-value function $Q_\theta(s_t, a)$ via a re-uploading QNN;
    Select action a_t using ϵ-greedy policy;
    Decompose A_t into control components $\{a_t^{tcl}, a_t^{price}, a_t^{ess}\}$;
    Apply control action A_t; ;
    Observe reward r_t and next state s_{t+1};
    Store transition (s_t, a_t, r_t, s_{t+1}) in replay buffer $\mathcal{D}$;
    Normalize s_{t+1} and encode it using $U_{enc}(S_{t+1}^*)$;
    Compute TD target according to Equation 21;
    Update Q-network parameters θ based on Equation 22;
    $s_i \leftarrow s_{i+1}$
  if *episode mod* $\zeta = 0$ **then**
    Update target network: $\theta' \leftarrow \theta$;
  Decay exploration rate ϵ;
Return trained Q-network Q_θ.

4.5. Training Process

Building upon the proposed QNN architecture, the training process enables the Q-EMS to learn an effective energy management policy through QRL, as illustrated in Figure 4. In this framework, the QNN serves as a value function approximator that estimates the expected long-term return associated with different control actions under a given system state. At each time step, the observed microgrid state is encoded into the quantum circuit and processed through successive encoding, variational, and entanglement layers. The measurement stage then extracts the expectation value of a selected quantum observable, which is

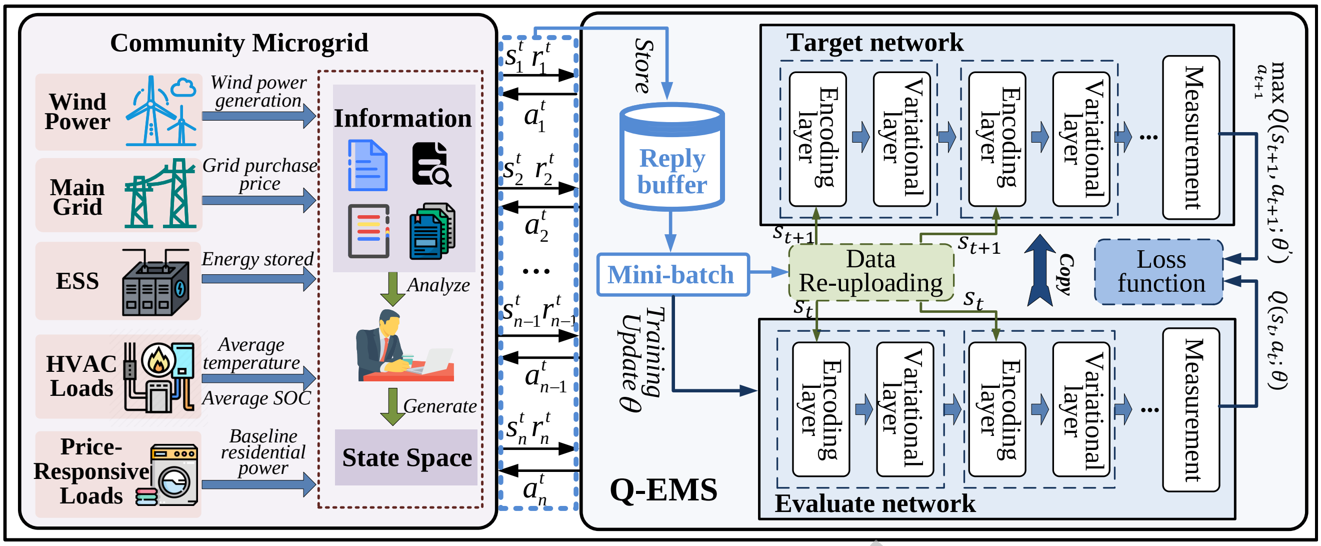


Figure 4: The process of interaction and training between QRL within Q-EMS and a real-world community microgrid environment.

interpreted as the estimated Q-value corresponding to the current decision context. The learning procedure follows the temporal-difference (TD) paradigm commonly adopted in Q-learning [28]. After executing an action and observing the immediate reward, the Q-value estimate is updated based on the observed transition. Specifically, the Q-value update is expressed as:

$$Q(S_t, a_t) = r_t + \gamma \max_{a' \in \mathcal{A}} Q'(S_{t+1}, a') \tag{21}$$

where r_t denotes the immediate reward obtained at time step t, γ is the discount factor, and $Q'(S_{t+1})$ represents the target Q-value associated with the subsequent system state.

To stabilize the learning process, a target network is introduced alongside the value network [6]. For each observed transition $\mathcal{P}_m = \{S_t, a_t, r_t, S_{t+1}\}$, the training objective is to minimize the TD error by optimizing the loss function:

$$\mathcal{L}(\theta) = (Q(S_t, a_t; \theta) - [r_t + \gamma \max_{a'} Q(S_{t+1}, a'; \theta')])^2 \tag{22}$$

where θ denote the trainable parameters of the QNN used for value estimation, and θ' represent the parameters of the target network, which are updated periodically. This formulation penalizes deviations between the current Q-value estimate and the corresponding TD target, thereby guiding the QNN toward a consistent value approximation. [33].

The parameters θ in the variational gates of the value network are updated through gradient descent with a learning rate of η, as expressed below:

$$\theta' = \theta - \eta \frac{\partial \mathcal{L}(\theta)}{\partial \theta} \tag{23}$$

where η denotes the learning rate. By decoupling the target network from the online updates, this training strategy mitigates instability arising from self-referential value estimation and improves convergence behavior.

Upon convergence, the trained QNN enables the Q-EMS to evaluate and select discrete control actions at each time step. These actions jointly govern multiple controllable components of the microgrid, including TCL operation levels, energy storage charging or discharging modes, and load shifting decisions in response to dynamic electricity prices. By coordinating decisions across thermal comfort regulation, energy storage utilization, and demand flexibility, the proposed QRL-based Q-EMS achieves adaptive and resilient energy management that balances economic efficiency with user comfort under varying operating conditions.

The detailed training procedure of Q-EMS is illustrated in Algorithm 1. The process starts with the initialization of the environment and parameters of QNN (lines 1-2). At each decision step, the observed microgrid state is first normalized and encoded into a quantum state (lines 6-7), which is subsequently processed by a re-uploading QNN to evaluate the corresponding action-value function (line 8). Based on the estimated Q-values, an action is selected according to an ϵ-greedy strategy and executed in the environment, leading to an immediate reward and a subsequent system state (lines 9-12). The resulting transition tuple (s_t, a_t, r_t, s_{t+1}) is then stored in the replay buffer $\mathcal{D}$ for experience replay (line 13). The QNN parameters are optimized by minimizing the TD loss, where the target value is computed using a target network that is updated periodically to enhance training stability (lines 14-17). This interaction and learning process continues throughout each episode. At the end of an episode, the parameters of the target network are synchronized with those of the evaluated Q-network at a fixed interval ζ, and the exploration rate ϵ is decayed accordingly (lines 19-20). After training converges, the learned Q-network Q_θ is obtained and used to support energy management decision-making in the Q-EMS (line 21).

5. Experimental Evaluation

In this section, we report the experimental evaluation of our Q-EMS by the comparison to established baselines.

Table 4
Key training hyperparameters of the Q-EMS.

Parameters	Value
Learning rate η	0.002
Batch-size M	200
Exploration probability ϵ	0.9
Target network updates frequency ζ	10
Discount factor γ	0.8
Size of replay buffer N_Δ	4000
Training iterations N_{iter}	500
Maximum capacity of ESS E^{max}	1000 kWh
Maximum power of wind power generation G^{max}	250 kW
The minimum of indoor temperature T^{min}	19 °C
The maximum of indoor temperature T^{max}	25 °C
Comfort penalty threshold κ	200
Maximum comfort reward magnitude ω	2
Weight factor between profit and comfort μ	1

5.1. Experimental Setup

The experimental configuration used to evaluate Q-EMS is described in terms of the implementation environment, data sources, evaluation settings, and baselines.

5.1.1. Experimental Settings

The experiments were implemented in Python 3 using TensorFlow Quantum (TFQ), which provides dedicated support for quantum machine learning and is well suited to QRL-based applications. All experiments were conducted on a workstation equipped with an NVIDIA GeForce RTX 4090 GPU and 24 GB of system memory. Table 4 summarizes the key hyperparameters of Q-EMS and the QNN training settings. The decision-making horizon was discretized into 24 time steps per day, corresponding to hourly control intervals. To reduce the influence of randomness and provide a more reliable comparison, we repeated all experiments five times and reported the averaged results.

Regarding the data sources, the wind generation profiles were derived from wind farm data in Finland [26]. The estimated levelized cost of wind energy, denoted by c_{gen}, was set to 3.2 cents/kWh based on a wind farm project in Finland [26]. The down-regulation and up-regulation prices were obtained from the open database of the Finnish power system operator[1], while the outdoor temperature data were collected from meteorological records in Helsinki[2]. On the demand side, the system includes 100 TCL units and 150 residential loads. Following the settings in [26], key household-specific parameters were sampled from normal distributions to capture heterogeneity in thermal dynamics and demand response behavior. Specifically, c_a^i, c_m^i, q_i, and p_i^{ac} were sampled from $\mathcal{N}(0.004, 0.0008)$, $\mathcal{N}(0.3, 0.004)$, $\mathcal{N}(0.0, 0.01)$, and $\mathcal{N}(1.5, 0.01)$, respectively. The price sensitivity coefficient β_i was sampled from $\mathcal{N}(0.4, 0.3)$.

[1] https://data.fingrid.fi/en/datasets?search=open-data-forms
[2] https://en.ilmatieteenlaitos.fi/download-observations

5.1.2. Baselines

The objective of the experimental evaluation is to examine whether the proposed QRL-based Q-EMS can achieve performance comparable to, or better than, advanced DRL methods, which already offer substantial improvements over conventional real-time energy management strategies. To this end, three representative baseline methods are selected for comparison:

- **Fixed:** This baseline adopts a rule-based control strategy with predefined decision rules that primarily prioritize user comfort, but lacks adaptive decision-making capability under dynamic conditions.
- **DQN:** This baseline is based on the DQN framework, which has been widely recognized as an effective approach for balancing user comfort and operational profitability in energy management systems [21]. It serves as a strong DRL benchmark for evaluating the relative performance of the proposed Q-EMS.
- **Improved SAC:** This baseline adopts an improved soft actor-critic (SAC) framework for microgrid energy management [16], which incorporates prioritized experience replay and an exploration-exploitation trade-off mechanism to enhance training efficiency.

5.2. Results

Experimental results are presented to evaluate the effectiveness of the proposed Q-EMS, with particular attention to training efficiency, economic performance, and indoor temperature regulation under different control strategies.

5.2.1. Training Convergence Comparison

The training convergence of the proposed Q-EMS is evaluated against DQN baselines with different network scales, as shown in Figure 5. Q-EMS adopts a compact 7-qubit QNN with three quantum circuit layers. In this architecture, the encoder introduces 21 input-dependent rotation angles, while the variational component applies three trainable single-qubit rotations to each qubit in each layer, resulting in 63 trainable rotation angles in total. Despite this compact structure, Q-EMS rapidly improves its training reward during the early training stage and reaches a stable high-reward level within approximately 200 episodes. This result indicates that the proposed QNN-based architecture can learn an effective energy management policy with relatively low parameter complexity.

For comparison, Figure 5(a) reports the training rewards of DQN models with one hidden layer, where the number of hidden neurons varies from 16 to 256. These models generally exhibit slower reward improvement and lower final reward levels, particularly when the hidden layer is small. This suggests that insufficient model capacity may limit action-value approximation and weaken reward convergence during training. Figure 5(b) presents the results of DQN models with two hidden layers. Compared with the one-hidden-layer models, the two-hidden-layer DQN baselines achieve higher

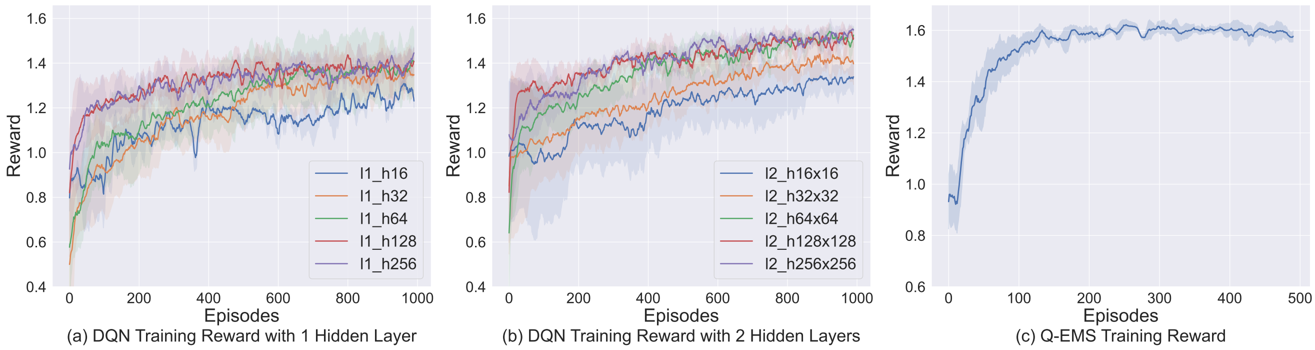


Figure 5: Training reward comparison between DQN baselines with different network architectures and Q-EMS: (a) DQN training reward with 1 hidden layer, (b) DQN training reward with 2 hidden layers, and (c) Q-EMS training reward.

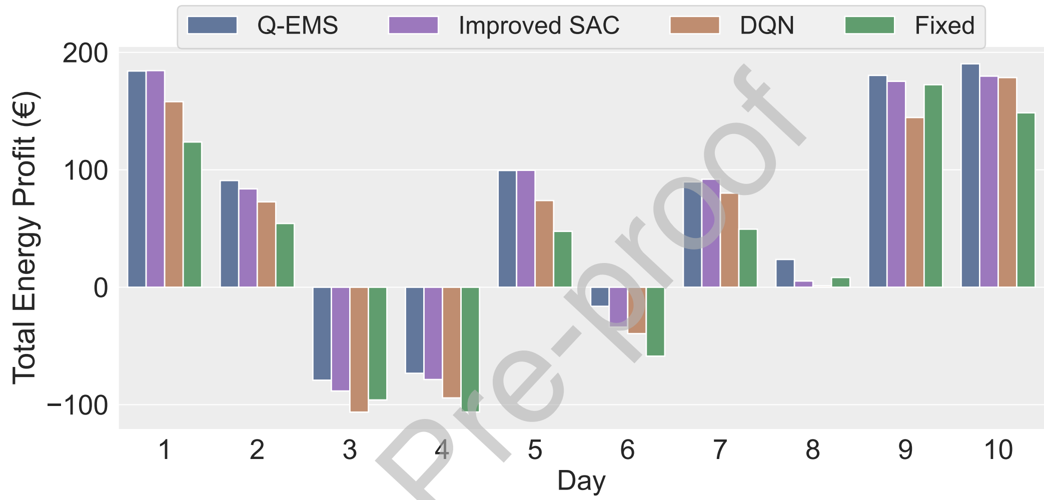


Figure 6: Comparison results of daily energy profits.

overall rewards, indicating that an appropriate increase in network depth can enhance the approximation capability of the classical value network. A reward level comparable to that of Q-EMS is observed when the hidden size reaches approximately 64×64. However, further increasing the network scale does not consistently improve training performance. For example, when the hidden size increases to 256×256, the reward improvement becomes marginal and even shows slight degradation compared with some smaller two-layer configurations. These results suggest that simply enlarging the classical network may increase optimization difficulty without guaranteeing better reward convergence. Overall, Q-EMS achieves competitive and stable training performance with substantially fewer trainable parameters, demonstrating favorable parameter efficiency for the considered energy management problem.

5.2.2. Performance Comparison

A comprehensive evaluation is conducted to assess the effectiveness of Q-EMS in terms of economic efficiency and user comfort, with the performance comparison carried out from the perspectives of total energy profit, energy transactions, and household indoor temperature.

Total Energy Profit. The primary objective of the proposed Q-EMS is to improve microgrid profitability while preserving user comfort. To evaluate its economic performance, Figure 6 compares the total energy profit over 10 consecutive testing days under four control strategies. Overall, Q-EMS consistently outperforms the Fixed strategy throughout the testing horizon, demonstrating its clear advantage over rule-based control. Compared with the DRL-based baselines, Q-EMS achieves higher profitability than DQN on most testing days and remains highly competitive with the improved SAC, outperforming it on several days, particularly Days 2, 3, 6, 8, and 10. Notably, under unfavorable operating conditions that lead to negative profits, Q-EMS reduces the magnitude of losses more effectively than the baseline methods, indicating stronger adaptability to challenging environments. Although daily profits vary with operating conditions, Q-EMS maintains a stable performance advantage across the evaluation period. These results demonstrate that Q-EMS can learn effective energy management policies and deliver superior long-term economic returns for community microgrids in dynamic environments.

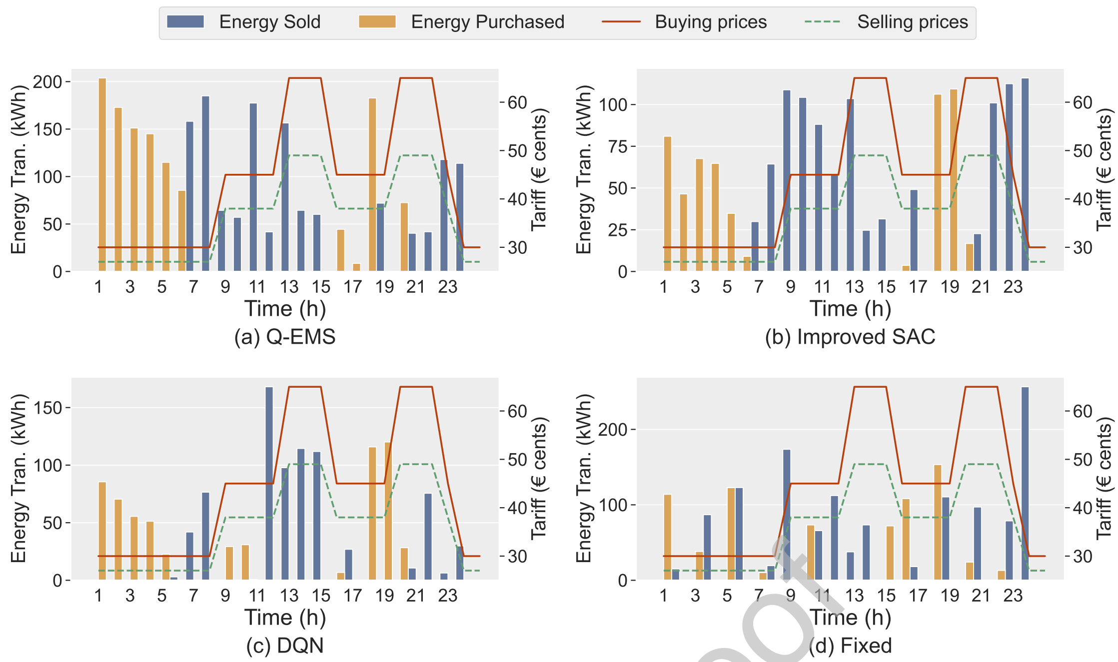


Figure 7: Energy transactions with the main grid under ToU tariffs with different energy management strategies: (a) Q-EMS, (b) Improved SAC, (c) DQN, and (d) Fixed.

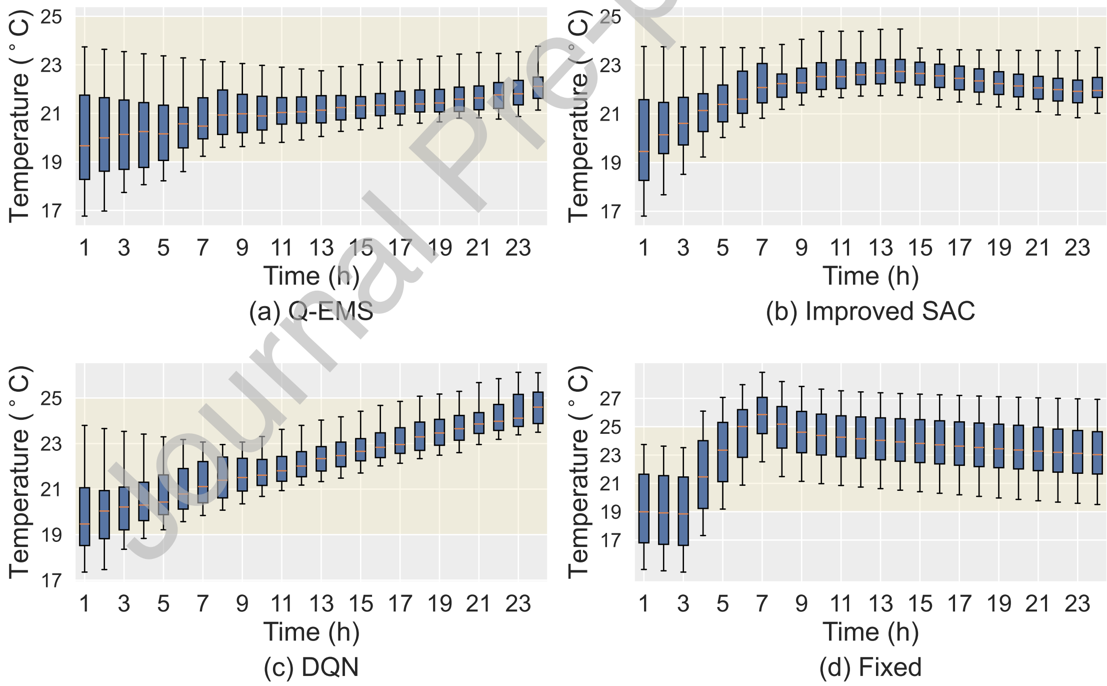


Figure 8: Household indoor temperature variations under different energy management strategies: (a) Q-EMS, (b) Improved SAC, (c) DQN, and (d) Fixed.

Energy Transactions. Efficient interaction with the main grid is essential for improving the economic performance of community microgrids. Figure 7 compares the electricity trading behaviors of Q-EMS and the three baseline methods under ToU tariffs on the eighth testing day. The Fixed strategy follows a relatively rigid trading pattern: it still purchases electricity during several high-price periods, such as Hours 15 and 17, while selling energy during low-price periods, such as Hours 1, 3, 5, 7, and 9. This indicates its limited adaptability to dynamic price signals. In contrast, Q-EMS and the DRL-based baselines, i.e., DQN and improved SAC, exhibit more price-aware trading behaviors by generally purchasing electricity during

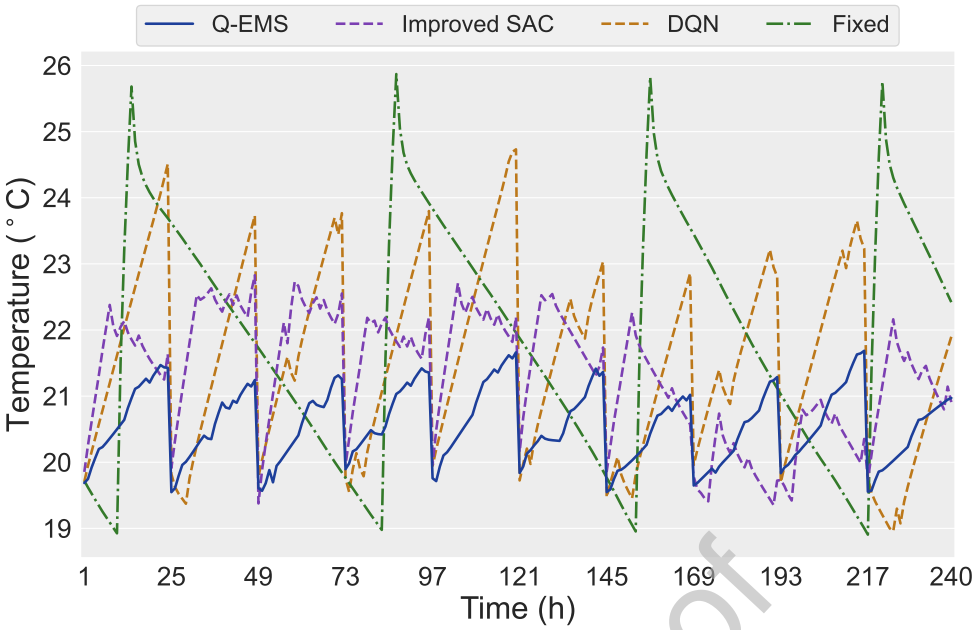


Figure 9: Comparison of ten-day average indoor temperature under Q-EMS, Improved SAC, DQN, and the Fixed strategy.

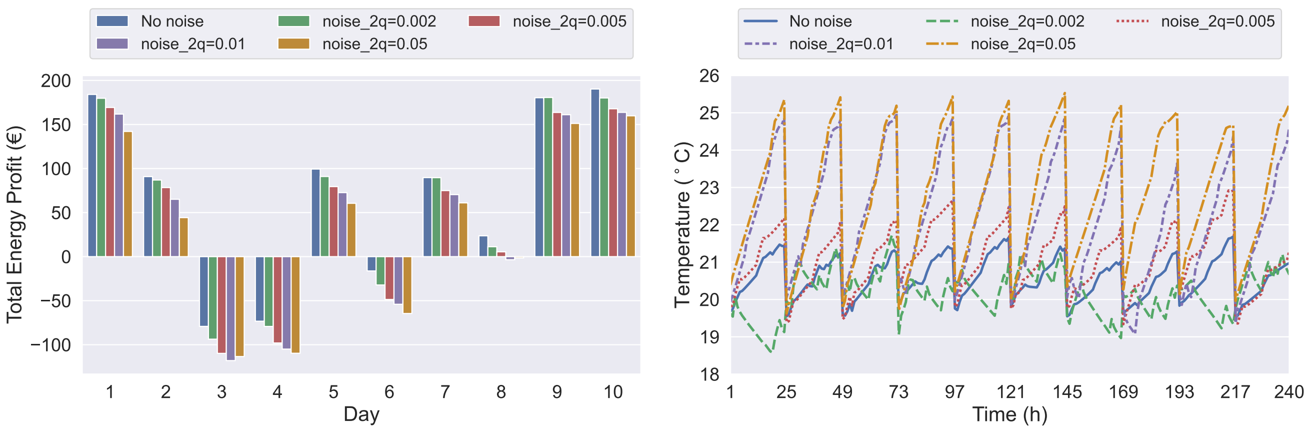
low-price periods and selling electricity during high-price periods. Among them, Q-EMS achieves a more prudent and economically effective trading pattern, with purchasing and selling decisions better aligned with ToU price variations. This behavior contributes to improved profit generation and demonstrates the capability of Q-EMS to perform cost-effective energy transactions under realistic pricing schemes.

Household Indoor Temperature. In addition to economic performance, the microgrid is required to maintain comfortable indoor thermal conditions across multiple households with heterogeneous thermal characteristics. Figure 8 presents the indoor temperature distributions over the first testing day under Q-EMS and the baseline methods. The results show that the Fixed strategy leads to pronounced temperature fluctuations and larger deviations from the desired comfort range. In contrast, Q-EMS and the DRL-based baselines maintain indoor temperatures within a narrower and more stable range throughout the day. Although DQN improves thermal regulation compared with the Fixed strategy, larger temperature variations are still observed compared with Q-EMS and the improved SAC. By adaptively adjusting energy allocation in response to real-time system states, Q-EMS enables more coordinated thermostatically controllable load operation and effectively limits deviations from the desired comfort range. These observations indicate that Q-EMS is capable of learning control policies that improve thermal stability while accommodating household-level diversity.

To further evaluate long-term temperature regulation performance, Figure 9 reports the average indoor temperature trajectories over ten testing days. Both Q-EMS and the DRL-based methods achieve better thermal regulation than the Fixed strategy, keeping indoor temperatures within the acceptable comfort bounds. However, differences can still be observed in the smoothness and stability of the temperature trajectories. DQN and the improved SAC tend to produce relatively coarse control adjustments, leading to more evident periodic temperature oscillations. In contrast, Q-EMS adjusts its control actions more smoothly according to the real-time system state, thereby maintaining thermal comfort while avoiding excessive corrective responses. Overall, these results indicate that Q-EMS achieves temperature regulation performance comparable to, or better than, the DRL-based methods while also improving energy efficiency. This further demonstrates the potential of QRL for supporting both effective temperature regulation and efficient energy management in advanced microgrid systems.

5.2.3. Noise Sensitivity Analysis

In the noisy intermediate-scale quantum (NISQ) era, quantum noise may affect the decision-making performance of QRL-based methods. To assess the robustness of Q-EMS under hardware-relevant noise, we conduct a depolarizing noise sensitivity analysis using TFQ and Cirq. In this experiment, we focus on two-qubit gate noise and insert a two-qubit depolarizing channel after each CZ gate. The two-qubit depolarizing strength is varied over $p_2 \in \{0, 0.002, 0.005, 0.01, 0.05\}$. Here, $p_2 = 0$ denotes the ideal noiseless case, while $p_2 = 0.002$, $p_2 = 0.005$, and $p_2 = 0.01$ correspond to error probabilities of 0.2%, 0.5%, and 1%, respectively. The setting $p_2 = 0.05$ is further included as a stress-test case under stronger noise. For each noise level, the Q-EMS policy is trained and evaluated under the same noise condition, and the corresponding results are shown in Figure 10.



(a) Total Energy Profit

(b) Average Temperature

Figure 10: Performance of Q-EMS under two-qubit depolarizing noise.

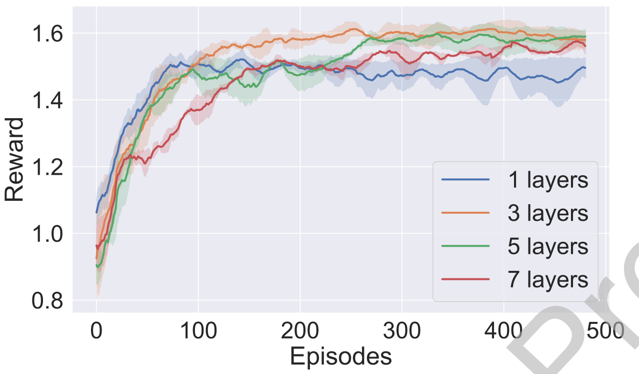


Figure 11: Reward convergence curves of QNNs with different circuit depths in the ablation study. The 1-layer QNN serves as the baseline without data re-uploading, while the 3-, 5-, and 7-layer QNNs perform data re-uploading across deeper quantum circuits.

As shown in Figure 10, Q-EMS maintains relatively stable performance under mild depolarizing noise. When $p_2 = 0.002$ and $p_2 = 0.005$, the total energy profit remains close to the noiseless case, while the average indoor temperature stays within the prescribed comfort range. This indicates that the proposed QRL-based policy can tolerate low-level two-qubit gate noise without substantial degradation. However, when the noise level increases to $p_2 = 0.01$ and $p_2 = 0.05$, Q-EMS exhibits more noticeable performance degradation, including reduced energy profit and larger indoor temperature oscillations. These results suggest that accumulated two-qubit gate noise can perturb action-value estimation and weaken control stability, although Q-EMS remains reasonably robust under mild noise conditions.

5.2.4. Ablation study

To evaluate the contribution of the data re-uploading architecture to the representational capability of the QNN in the proposed Q-EMS framework, an ablation study is conducted by varying the number of quantum circuit layers. The one-layer QNN encodes the classical state only once and serves as the baseline without re-uploading, whereas the 3-, 5-, and 7-layer QNNs introduce 2, 4, and 6 additional data re-uploading blocks, respectively, each followed by variational quantum transformations. The corresponding reward convergence curves are shown in Figure 11.

The results show that the QNN without data re-uploading improves rapidly in the early training stage, mainly due to its compact structure and smaller parameter space. However, its reward soon saturates at a lower level, indicating limited expressive capability in modeling complex state-action relationships. In contrast, QNNs with data re-uploading achieve higher converged rewards, demonstrating that repeated state encoding and variational transformations can enrich quantum representations and improve decision-making capability. Among all configurations, the 3-layer QNN achieves the best balance between convergence speed and final performance. Although the 5- and 7-layer QNNs also reach competitive rewards, their training curves exhibit stronger fluctuations, suggesting that excessive re-uploading may reduce training stability without yielding commensurate performance gains. Therefore, the re-uploading depth should be carefully selected to balance model capacity, training stability, and convergence efficiency.

6. Conclusion

In this paper, we proposed Q-EMS, a QRL framework for microgrid energy management. The framework integrates a carefully designed quantum neural network architecture to effectively capture system dynamics and support adaptive decision-making. By exploiting the expressive capability of quantum models, Q-EMS achieves improved energy efficiency and cost performance while maintaining user comfort in community microgrids. The design and implementation of the proposed approach were presented in detail, and experimental results demonstrate that Q-EMS attains performance comparable to, and in some cases exceeding, that of state-of-the-art deep reinforcement learning methods. Future work will focus on extending Q-EMS to incorporate additional renewable energy sources and energy storage technologies, as well as further exploring the role of quantum reinforcement learning in enhancing clean energy utilization in modern microgrid systems.

CRediT authorship contribution statement

Yan Gu: Investigation, Methodology, Software, Writing – original draft. **Huiru Yan**: Methodology, Validation, Writing – review & editing. **Qingle Wang**: Conceptualization, Writing – review & editing. **Cong Liu**: Conceptualization, Writing – review & editing. **Cheng Liu**: Methodology, Writing – review & editing. **Long Cheng**: Conceptualization, Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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